JUNIOR SCHOOL LEARNING HUB

GRADE 8 MATHEMATICS — SCHEME OF WORK — TERM 3 2027

School: _____ Teacher: _____ TSC No.: _____

CBC planning format | Lesson outcomes are organised into Knowledge, Skills and Attitudes. Learning experiences retain the activities in the supplied Grade 8 scheme.

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
2	1	4.0: Geometry	4.3: Scale Drawing - Representation of length to given scale	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define scale and its purpose **Skills:** Determine scale from given measurements **Attitudes:** Show understanding of proportion	**In groups, learners are guided to:** • Compare sizes of objects and their representations • Discuss need for scale in drawings • Measure actual dimensions • Choose appropriate scale for representations • Calculate scale from given information • Express scale in different forms	Why do we need scale when drawing large objects?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Tape measure - Pencil - Drawing paper	- Observation - Oral questions - Practical tasks	
2	2	4.0: Geometry	4.3: Scale Drawing - Converting actual length to scale length	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for converting actual length to scale length **Skills:** Convert actual measurements to scale measurements accurately **Attitudes:** Demonstrate computational skills	**In groups, learners are guided to:** • Apply given scales to convert measurements • Complete tables converting actual to scale lengths • Calculate scale lengths using various scales • Work with different units • Practice systematic conversions	How do we calculate scale length from actual length?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Conversion tables - Pencil	- Observation - Written assignments - Problem-solving	
2	3	4.0: Geometry	4.3: Scale Drawing - Converting actual length to scale length	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for converting actual length to scale length **Skills:** Convert actual measurements to scale measurements accurately **Attitudes:** Demonstrate computational skills	**In groups, learners are guided to:** • Apply given scales to convert measurements • Complete tables converting actual to scale lengths • Calculate scale lengths using various scales • Work with different units • Practice systematic conversions	How do we calculate scale length from actual length?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Conversion tables - Pencil	- Observation - Written assignments - Problem-solving	
2	4	4.0: Geometry	4.3: Scale Drawing - Converting scale length to actual length	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of converting scale to actual measurements **Skills:** Convert scale measurements to actual measurements accurately **Attitudes:** Show systematic calculation approach	**In groups, learners are guided to:** • Measure lengths on scale diagrams • Use given scales to find actual lengths • Calculate actual distances • Work with different unit conversions • Practice reverse calculations	How do we find real dimensions from scale drawings?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Scale drawings - Pencil	- Observation - Written tests - Practical tasks	
2	5	4.0: Geometry	4.3: Scale Drawing - Interpreting linear scales in statement form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define linear scale **Skills:** Interpret scale markings and express in statement form **Attitudes:** Show understanding of scale representation	**In groups, learners are guided to:** • Examine linear scales on maps and plans • Measure length of scale markings • Determine what distance each unit represents • Practice with different linear scales • Express linear scales using words	How do linear scales differ from numerical scales?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Maps with linear scales - Ruler - Pencil - Sample plans	- Observation - Oral questions - Written assignments	
3	1	4.0: Geometry	4.3: Scale Drawing - Writing linear scales in statement form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall the format for writing scales in statement form **Skills:** Express scales in statement form clearly and accurately **Attitudes:** Demonstrate understanding of scale notation	**In groups, learners are guided to:** • Express given scales in statement form • Write statements using proper format • Practice with scales showing various divisions • Convert linear scales to statements • Discuss advantages of statement form	Why is statement form useful for describing scales?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Linear scale examples - Pencil - Drawing paper	- Observation - Written tests - Practical tasks	
3	2	4.0: Geometry	4.3: Scale Drawing - Interpreting linear scales in ratio form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define ratio form of scales **Skills:** Convert measurements to same units and express scales as ratios **Attitudes:** Show understanding of proportional relationships	**In groups, learners are guided to:** • Convert scales ensuring same units • Express scales as ratios • Practice unit conversions before writing ratios • Work with various scales • Understand ratios have no units	What does a scale ratio tell us about a drawing?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Conversion charts - Pencil	- Observation - Problem-solving - Oral questions	
3	3	4.0: Geometry	4.3: Scale Drawing - Interpreting linear scales in ratio form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define ratio form of scales **Skills:** Convert measurements to same units and express scales as ratios **Attitudes:** Show understanding of proportional relationships	**In groups, learners are guided to:** • Convert scales ensuring same units • Express scales as ratios • Practice unit conversions before writing ratios • Work with various scales • Understand ratios have no units	What does a scale ratio tell us about a drawing?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Conversion charts - Pencil	- Observation - Problem-solving - Oral questions	
3	4	4.0: Geometry	4.3: Scale Drawing - Writing linear scales in ratio form	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the requirements for writing scales in ratio form **Skills:** Write scales in ratio form correctly without units **Attitudes:** Demonstrate accuracy in conversions	**In groups, learners are guided to:** • Complete tables converting statement to ratio form • Convert scales with various measurements • Write map scales in ratio form • Calculate ratios for different scenarios • Practice systematic conversions	How do we ensure accuracy when converting to ratio form?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Conversion tables - Pencil - Practice worksheets	- Observation - Written assignments - Problem-solving	
3	5	4.0: Geometry	4.3: Scale Drawing - Converting scale from statement to ratio form	**At the end of the lesson, the learner should be able to:** **Knowledge:** List the steps for converting statement to ratio form **Skills:** Convert statement form scales to ratio form systematically **Attitudes:** Show computational proficiency	**In groups, learners are guided to:** • Convert statement scales to ratio form • Practice with different unit combinations • Apply systematic conversion process • Work with plans and maps • Verify conversions	What steps ensure correct conversion from statement to ratio?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Unit conversion chart - Pencil	- Observation - Written tests - Practical tasks	
4	1	4.0: Geometry	4.3: Scale Drawing - Converting scale from ratio to statement form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of converting ratio to statement form **Skills:** Convert ratio form scales to statement form using appropriate units **Attitudes:** Demonstrate understanding of both forms	**In groups, learners are guided to:** • Convert ratio scales to statement form • Determine appropriate units for actual measurements • Express scales clearly in words • Practice with various ratio scales • Choose suitable units for statements	How do we choose appropriate units in statement form?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Calculator - Ruler - Pencil	- Observation - Problem-solving - Oral questions	

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4	2	4.0: Geometry	4.3: Scale Drawing - Converting scale from ratio to statement form	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the process of converting ratio to statement form **Skills:** Convert ratio form scales to statement form using appropriate units **Attitudes:** Demonstrate understanding of both forms	**In groups, learners are guided to:** • Convert ratio scales to statement form • Determine appropriate units for actual measurements • Express scales clearly in words • Practice with various ratio scales • Choose suitable units for statements	How do we choose appropriate units in statement form?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Calculator - Ruler - Pencil	- Observation - Problem-solving - Oral questions	
4	3	4.0: Geometry	4.3: Scale Drawing - Making scale drawings with calculations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify dimensions needed for scale drawings **Skills:** Calculate scale lengths and make accurate scale drawings **Attitudes:** Show precision in measurements and drawing	**In groups, learners are guided to:** • Calculate scale lengths before drawing • Make accurate scale drawings of various shapes • Apply appropriate scales • Measure and verify dimensions • Calculate areas from scale drawings	Why must we calculate scale lengths before drawing?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Pencil - Calculator - Drawing paper	- Observation - Practical construction - Written tests	
4	4	4.0: Geometry	4.3: Scale Drawing - Scale drawings with distance calculations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Recall how to measure distances on drawings **Skills:** Make scale drawings involving multiple distances and calculate actual distances **Attitudes:** Show systematic approach to problem-solving	**In groups, learners are guided to:** • Make scale drawings involving multiple points • Use suitable scales for given distances • Measure lengths on scale drawings • Calculate actual distances from drawings • Apply geometric principles where needed • Verify measurements	How do scale drawings help solve distance problems?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Pair of compasses - Calculator - Graph paper	- Observation - Practical tasks - Problem-solving	
4	5	4.0: Geometry	4.3: Scale Drawing - Using maps and demonstrating scale	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify scales on actual maps **Skills:** Read scales from maps and measure distances accurately **Attitudes:** Appreciate real-world applications of scale	**In groups, learners are guided to:** • Examine maps in atlas • Identify and read map scales • Measure distances between locations • Calculate actual distances using scale • Compare different maps with different scales • Discuss map features	How does scale choice affect what we can show on a map?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Maps - Ruler - Calculator - Digital resources	- Observation - Practical measurement - Oral questions	
5	1	4.0: Geometry	4.3: Scale Drawing - Application problems with scale	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify given information in scale problems **Skills:** Solve complex problems involving scale, area, volume, time and speed **Attitudes:** Show advanced problem-solving skills	**In groups, learners are guided to:** • Solve problems involving height and scale • Find scales used in given scenarios • Calculate areas from scale diagrams • Determine time and speed using map scales • Work with various measurement scenarios • Apply multiple concepts together	How do professionals use scale in their work?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Problem cards - Reference materials	- Observation - Problem-solving - Written tests	
5	2	4.0: Geometry	4.3: Scale Drawing - Application problems with scale	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify given information in scale problems **Skills:** Solve complex problems involving scale, area, volume, time and speed **Attitudes:** Show advanced problem-solving skills	**In groups, learners are guided to:** • Solve problems involving height and scale • Find scales used in given scenarios • Calculate areas from scale diagrams • Determine time and speed using map scales • Work with various measurement scenarios • Apply multiple concepts together	How do professionals use scale in their work?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Problem cards - Reference materials	- Observation - Problem-solving - Written tests	
5	3	4.0: Geometry	4.3: Scale Drawing - Using ICT for scale and maps	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe how digital maps use scale **Skills:** Use digital devices to display maps and demonstrate zoom functions **Attitudes:** Show digital literacy in geography context	**In groups, learners are guided to:** • Access digital maps on devices • Use zoom function to change scale • Observe how scale changes with zoom level • Measure distances on digital maps • Compare scale indicators on digital and paper maps • Discuss advantages of digital tools	How does zooming affect the scale of a digital map?	- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Digital devices (tablets/computers) - Internet access - Digital mapping software - Projector	- Observation - Practical demonstration - Oral questions	
5	4	4.0: Geometry	4.4: Common Solids - Identifying common solids from environment	**At the end of the lesson, the learner should be able to:** **Knowledge:** Name common solids: cubes, cuboids, cylinders, pyramids and cones **Skills:** Classify solids by their properties **Attitudes:** Show awareness of geometric shapes in environment	**In groups, learners are guided to:** • Collect objects from environment • Group objects by shape categories • Identify properties of each solid type • Discuss examples in daily life • Create display of classified solids	Where do we see these solids in our daily lives?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Collection of solid objects - Models of solids - Pictures of buildings - Digital images	- Observation - Practical classification - Oral questions	
5	5	4.0: Geometry	4.4: Common Solids - Properties of solids (faces, edges, vertices)	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define faces, edges and vertices **Skills:** Identify and count faces, edges and vertices of given solids **Attitudes:** Show understanding of 3D properties	**In groups, learners are guided to:** • Examine labeled solids • Name all faces of solids • Identify all edges • Locate all vertices • Practice with different solids • Record properties systematically	How do faces, edges and vertices define a solid?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Models of solids - Ruler - Labels - Worksheet	- Observation - Written assignments - Practical identification	
6	1	4.0: Geometry	4.4: Common Solids - Sketching nets of cubes	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define the term "net" of a solid **Skills:** Sketch nets of closed and open cubes **Attitudes:** Demonstrate spatial visualization	**In groups, learners are guided to:** • Label cube vertices • Cut cube along specified edges • Lay out faces on flat surface • Sketch net showing all faces for closed cube • Sketch net showing appropriate faces for open cube • Identify different possible net arrangements	How does a 3D cube transform into a 2D net?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Scissors/razor blade - Ruler - Pencil - Plain paper	- Observation - Practical construction - Peer assessment	
6	2	4.0: Geometry	4.4: Common Solids - Sketching nets of cubes	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define the term "net" of a solid **Skills:** Sketch nets of closed and open cubes **Attitudes:** Demonstrate spatial visualization	**In groups, learners are guided to:** • Label cube vertices • Cut cube along specified edges • Lay out faces on flat surface • Sketch net showing all faces for closed cube • Sketch net showing appropriate faces for open cube • Identify different possible net arrangements	How does a 3D cube transform into a 2D net?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Scissors/razor blade - Ruler - Pencil - Plain paper	- Observation - Practical construction - Peer assessment	

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6	3	4.0: Geometry	4.4: Common Solids - Sketching nets of cuboids	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify faces of cuboids **Skills:** Sketch nets of closed and open cuboids **Attitudes:** Show accuracy in net construction	**In groups, learners are guided to:** • Label cuboid vertices • Cut along specified edges • Spread faces on flat surface • Sketch net with all faces for closed cuboid • Sketch net with appropriate faces for open cuboid • Identify pairs of equal faces	How does the net of a cuboid differ from that of a cube?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cuboids - Scissors/razor blade - Ruler - Pencil - Grid paper	- Observation - Practical tasks - Written tests	
6	4	4.0: Geometry	4.4: Common Solids - Sketching nets of cylinders	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify components of cylinder nets **Skills:** Sketch nets of closed and open cylinders **Attitudes:** Understand curved surface as rectangle	**In groups, learners are guided to:** • Cut cylinder to remove bases • Cut along height to open curved surface • Observe curved surface forms rectangle • Note rectangle dimensions relate to circumference • Sketch nets for closed and open cylinders • Label components	Why does the curved surface become a rectangle?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Scissors/razor blade - Ruler - Pair of compasses - Pencil	- Observation - Practical construction - Oral questions	
6	5	4.0: Geometry	4.4: Common Solids - Sketching nets of pyramids	**At the end of the lesson, the learner should be able to:** **Knowledge:** Describe components of pyramid nets **Skills:** Sketch nets of pyramids with different bases **Attitudes:** Show precision in drawing nets	**In groups, learners are guided to:** • Label pyramid vertices • Cut along slant edges • Lay faces on flat surface • Sketch net showing base and triangular faces • Ensure triangular faces connect to base edges • Practice with different base dimensions	How many triangular faces does a square-based pyramid have?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model pyramids - Scissors/razor blade - Ruler - Pencil - Drawing paper	- Observation - Practical tasks - Peer review	
7	1	4.0: Geometry	4.4: Common Solids - Sketching nets of cones	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify components of cone nets **Skills:** Sketch nets of cones showing sector shape **Attitudes:** Appreciate relationship between arc and circumference	**In groups, learners are guided to:** • Cut base from cone • Cut curved surface along slant height • Observe curved surface forms sector • Note relationship between arc length and base circumference • Sketch net showing circle and sector • Label components	Why does the cone's curved surface form a sector?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones - Scissors/razor blade - Protractor - Pair of compasses - Pencil	- Observation - Practical construction - Written assignments	
7	2	4.0: Geometry	4.4: Common Solids - Sketching nets of cones	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify components of cone nets **Skills:** Sketch nets of cones showing sector shape **Attitudes:** Appreciate relationship between arc and circumference	**In groups, learners are guided to:** • Cut base from cone • Cut curved surface along slant height • Observe curved surface forms sector • Note relationship between arc length and base circumference • Sketch net showing circle and sector • Label components	Why does the cone's curved surface form a sector?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones - Scissors/razor blade - Protractor - Pair of compasses - Pencil	- Observation - Practical construction - Written assignments	
7	3	4.0: Geometry	4.4: Common Solids - Matching solids to nets and vice versa	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify solids from their nets **Skills:** Match given solids to correct nets **Attitudes:** Demonstrate spatial reasoning	**In groups, learners are guided to:** • Match various solids to their nets • Identify which solid each net will form • Draw solid that corresponds to given net • Practice visualizing 3D from 2D • Sketch nets for solids with various dimensions	How can we visualize the solid from looking at its net?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Various nets - Model solids - Ruler - Pencil - Matching cards	- Observation - Practical matching - Problem-solving	
7	4	4.0: Geometry	4.4: Common Solids - Surface area of cubes from nets	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for surface area of cube **Skills:** Calculate total surface area of cube from its net **Attitudes:** Show systematic calculation approach	**In groups, learners are guided to:** • Measure sides of cube • Sketch net of cube • Calculate area of one face • Multiply by number of faces • Practice with cubes of different dimensions • Verify by drawing net and calculating	How does knowing one side help find total surface area?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Ruler - Calculator - Pencil - Net templates	- Observation - Written tests - Problem-solving	
7	5	4.0: Geometry	4.4: Common Solids - Surface area of cuboids from nets	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for surface area of cuboid **Skills:** Calculate total surface area of cuboid from nets **Attitudes:** Show organized calculation method	**In groups, learners are guided to:** • Draw net of cuboid with given dimensions • Calculate areas of different faces • Identify pairs of equal faces • Add all areas to find total • Practice with various dimensions • Verify calculations	Why do we calculate surface area in pairs for cuboids?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cuboids - Ruler - Calculator - Grid paper - Pencil	- Observation - Written assignments - Practical calculations	
8	1	4.0: Geometry	4.4: Common Solids - Surface area of cylinders from nets	**At the end of the lesson, the learner should be able to:** **Knowledge:** State components of cylinder surface area **Skills:** Calculate total surface area of cylinder from nets **Attitudes:** Demonstrate formula application	**In groups, learners are guided to:** • Identify net components • Calculate area of circular faces • Find rectangle dimensions using circumference • Calculate rectangular area • Add areas for total surface area • Practice with different dimensions	How is the circumference used in finding surface area?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Ruler - Calculator - Pair of compasses - Formula chart	- Observation - Problem-solving - Written tests	
8	2	4.0: Geometry	4.4: Common Solids - Surface area of cylinders from nets	**At the end of the lesson, the learner should be able to:** **Knowledge:** State components of cylinder surface area **Skills:** Calculate total surface area of cylinder from nets **Attitudes:** Demonstrate formula application	**In groups, learners are guided to:** • Identify net components • Calculate area of circular faces • Find rectangle dimensions using circumference • Calculate rectangular area • Add areas for total surface area • Practice with different dimensions	How is the circumference used in finding surface area?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Ruler - Calculator - Pair of compasses - Formula chart	- Observation - Problem-solving - Written tests	
8	3	4.0: Geometry	4.4: Common Solids - Surface area of pyramids from nets	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify components of pyramid surface area **Skills:** Calculate total surface area of pyramid from nets **Attitudes:** Show systematic approach to complex calculations	**In groups, learners are guided to:** • Draw net showing base and triangular faces • Calculate base area • Calculate area of each triangular face • Add base area to sum of triangular areas • Practice with different dimensions • Verify calculations	How do we find the slant height if not given?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model pyramids - Ruler - Calculator - Pencil - Net templates	- Observation - Written assignments - Problem-solving	

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8	4	4.0: Geometry	4.4: Common Solids - Surface area of cones and distance on surfaces	**At the end of the lesson, the learner should be able to:** **Knowledge:** State formula for surface area of cone **Skills:** Calculate surface area of cone from net and determine shortest distances on solid surfaces **Attitudes:** Show advanced spatial reasoning	**In groups, learners are guided to:** • Identify cone net components • Calculate circular base area • Calculate sector area using given angle • Find total surface area • Open cuboid into net to find paths between points • Measure distances along net surface	How does opening a solid help find distances on its surface?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones and cuboids - Protractor - Calculator - String - Scissors	- Observation - Problem-solving - Practical tasks	
8	5	4.0: Geometry	4.4: Common Solids - Surface area of cones and distance on surfaces	**At the end of the lesson, the learner should be able to:** **Knowledge:** State formula for surface area of cone **Skills:** Calculate surface area of cone from net and determine shortest distances on solid surfaces **Attitudes:** Show advanced spatial reasoning	**In groups, learners are guided to:** • Identify cone net components • Calculate circular base area • Calculate sector area using given angle • Find total surface area • Open cuboid into net to find paths between points • Measure distances along net surface	How does opening a solid help find distances on its surface?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones and cuboids - Protractor - Calculator - String - Scissors	- Observation - Problem-solving - Practical tasks	
9	1	4.0: Geometry	4.4: Common Solids - Making models of hollow solids (cubes and cuboids)	**At the end of the lesson, the learner should be able to:** **Knowledge:** List steps for making hollow models **Skills:** Construct hollow cube and cuboid models from nets **Attitudes:** Show craftsmanship in model making	**In groups, learners are guided to:** • Draw nets accurately on manila paper • Include flaps for joining faces • Cut out nets carefully • Fold along marked lines • Paste flaps to form hollow solids • Display completed models	Why do we need flaps when making hollow models?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Manila paper - Ruler - Pencil - Scissors - Glue/paste - Colored markers	- Observation - Practical construction - Peer assessment	
9	2	4.0: Geometry	4.4: Common Solids - Making models of cylinders, cones and pyramids	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain steps for making different hollow models **Skills:** Construct hollow cylinder, cone and pyramid models **Attitudes:** Show precision and craftsmanship	**In groups, learners are guided to:** • Draw cylinder net with calculated dimensions • Construct cone net with appropriate sector angle • Draw pyramid net with correct measurements • Cut and fold to form solids • Paste edges to complete models • Display and compare models	How do we ensure the cylinder's rectangle matches the circle?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Manila paper - Pair of compasses - Protractor - Ruler - Scissors - Glue	- Observation - Practical construction - Written reflection	
9	3	4.0: Geometry	4.4: Common Solids - Using IT devices and drawing technology	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify technology tools for learning about solids **Skills:** Use technology to explore and draw solids and nets **Attitudes:** Appreciate technology in mathematics learning	**In groups, learners are guided to:** • Watch educational videos about solids • Use software to draw 3D shapes • Explore rotating solids digitally • Practice drawing nets using technology • Use apps to visualize net folding • Share digital creations	How does technology enhance our understanding of 3D shapes?	- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Computers/tablets - Internet access - GeoGebra software - Projector - 3D modeling apps	- Observation - Digital portfolio - Oral presentation - Peer evaluation	
9	4	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Collecting data and drawing bar graphs	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define bar graph and identify its components **Skills:** Collect data from own experiences and draw bar graphs with suitable scale **Attitudes:** Appreciate the use of graphs in presenting data	**In groups, learners are guided to:** • Collect data from class members on given characteristics • Fill data in tables • Choose suitable scale for collected data • Draw bar graphs to represent collected data • Compare graphs with other groups • Discuss components of bar graphs	How can we represent collected data visually?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Ruler - Graph paper - Pencil - Data collection sheets	- Observation - Practical tasks - Oral questions	
9	5	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Drawing bar graphs with suitable scale 5.1: Data Presentation and Interpretation - Interpreting bar graphs	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the steps for drawing bar graphs **Skills:** Draw bar graphs with appropriate scales for different data sets **Attitudes:** Show accuracy in graph construction	**In groups, learners are guided to:** • Choose uniform width for bars • Select uniform gaps between bars • Choose suitable scale for vertical axis • Calculate heights of bars according to scale • Draw bars accurately • Label axes properly • Practice with various data sets	How do we choose an appropriate scale for a bar graph?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Graph paper - Ruler - Pencil - Calculator - Data tables - Sample bar graphs - Question sheets	- Observation - Practical construction - Written assignments	
10	1	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Drawing line graphs	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define line graph and state its uses **Skills:** Draw line graphs from given data **Attitudes:** Appreciate line graphs for showing trends	**In groups, learners are guided to:** • Choose suitable scale for x-axis • Choose suitable scale for y-axis • Plot points from table of values • Join plotted points using straight lines • Label axes appropriately • Practice drawing line graphs for different data sets	When is it appropriate to use a line graph instead of a bar graph?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Graph paper - Ruler - Pencil - Calculator - Data tables	- Observation - Practical construction - Peer assessment	
10	2	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Interpreting line graphs 5.1: Data Presentation and Interpretation - Identifying mode of discrete data	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain how to read values from line graphs **Skills:** Interpret line graphs to extract information **Attitudes:** Show analytical skills in reading trends	**In groups, learners are guided to:** • Read values at specific points on graph • Identify highest and lowest values • Determine trends from line graphs • Calculate totals from graph data • Answer questions based on line graphs • Discuss patterns observed	How do line graphs help us see changes over time?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Sample line graphs - Ruler - Pencil - Question sheets - Number cards - Exercise books - Data sets	- Observation - Written tests - Problem-solving	
10	3	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Calculating mean of discrete data 5.1: Data Presentation and Interpretation - Working out averages from different sets	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for calculating mean **Skills:** Calculate the mean of discrete data sets accurately **Attitudes:** Show systematic approach in calculations	**In groups, learners are guided to:** • Note down values from group members • Add all values in data set • Count number of values • Divide sum by number of values • Calculate mean for various data sets • Verify calculations • Practice with different contexts	How does the mean represent the average of a data set?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Calculator - Pencil - Exercise books - Data sets - Problem cards	- Observation - Written tests - Problem-solving	
10	4	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Determining median of discrete data	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define median and explain the process of finding it **Skills:** Determine the median of discrete data for odd and even sets **Attitudes:** Show systematic approach in finding median	**In groups, learners are guided to:** • Arrange data in ascending or descending order • Identify middle value for odd sets • Calculate median for even sets by averaging two middle values • Practice finding median for various data sets • Compare median with mode and mean • Discuss applications	Why must data be arranged in order before finding the median?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Number cards - Pencil - Exercise books - Calculator	- Observation - Oral questions - Written tests	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
10	5	5.0: Data Handling and Probability	5.1: Data Presentation and Interpretation - Using IT for data presentation and calculations	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify IT tools for creating graphs **Skills:** Use technology to create bar graphs and line graphs and calculate mean, mode and median **Attitudes:** Appreciate technology in data handling	**In groups, learners are guided to:** • Use spreadsheet software to enter data • Create bar graphs using software • Create line graphs using software • Use formulas to calculate mean • Use functions to find mode and median • Compare manual and digital methods • Present findings digitally	How does technology make data presentation and analysis easier?	- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Computers/tablets - Spreadsheet software - Internet access - Projector - Data sets	- Observation - Digital portfolio - Practical demonstration - Peer evaluation	
11	1	5.0: Data Handling and Probability	5.2: Probability - Identifying events involving chance in real life	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define chance and probability **Skills:** Identify events involving chance in daily life **Attitudes:** Show awareness of probability in real situations	**In groups, learners are guided to:** • Discuss possibilities in various scenarios • Identify chance events in sports • Recognize chance in weather predictions • Discuss chance in games • List daily events involving chance • Share observations with class	What is chance and where do we encounter it in daily life?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Pictures of chance events - Pencil - Chart paper - Real-life scenario cards	- Observation - Oral questions - Class discussion	
11	2	5.0: Data Handling and Probability	5.2: Probability - Identifying events involving chance in real life	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define chance and probability **Skills:** Identify events involving chance in daily life **Attitudes:** Show awareness of probability in real situations	**In groups, learners are guided to:** • Discuss possibilities in various scenarios • Identify chance events in sports • Recognize chance in weather predictions • Discuss chance in games • List daily events involving chance • Share observations with class	What is chance and where do we encounter it in daily life?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Pictures of chance events - Pencil - Chart paper - Real-life scenario cards	- Observation - Oral questions - Class discussion	
11	3	5.0: Data Handling and Probability	5.2: Probability - Discussing likely and unlikely events	**At the end of the lesson, the learner should be able to:** **Knowledge:** List the likelihood scale terms: impossible, unlikely, equally likely, likely, certain **Skills:** Classify events as impossible, unlikely, equally likely, likely or certain **Attitudes:** Show critical thinking in analyzing probability	**In groups, learners are guided to:** • Examine likelihood scale • Discuss meaning of each term • Classify statements using likelihood terms • Identify impossible events • Identify certain events • Distinguish between likely and unlikely • Practice with various statements	How do we describe the likelihood of different events happening?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Likelihood scale chart - Event cards - Pencil - Exercise books	- Observation - Oral questions - Written assignments	
11	4	5.0: Data Handling and Probability	5.2: Probability - Performing chance experiments	**At the end of the lesson, the learner should be able to:** **Knowledge:** Define chance experiment **Skills:** Perform chance experiments such as flipping coins, tossing dice, and drawing objects **Attitudes:** Show interest in hands-on probability activities	**In groups, learners are guided to:** • Obtain coins and flip them • Toss dice and record outcomes • Draw colored balls or beads from bags • Use spinners and record results • Record outcomes from experiments • Compare results with other groups • Discuss patterns observed	What are the possible outcomes when we perform chance experiments?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Coins - Dice - Colored balls/beads - Bags - Spinners - Recording sheets	- Observation - Practical tasks - Oral questions	
11	5	5.0: Data Handling and Probability	5.2: Probability - Writing experimental probability outcomes	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the concept of experimental probability **Skills:** Write all possible outcomes from chance experiments **Attitudes:** Demonstrate systematic recording of outcomes	**In groups, learners are guided to:** • List possible outcomes from coin toss • Write outcomes from die roll • Determine outcomes from spinners • List outcomes from drawing objects • Form combinations of outcomes • Record outcomes systematically • Share findings with class	How do we list all possible outcomes from an experiment?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Coins - Dice - Number cards - Pencil - Exercise books	- Observation - Written tests - Problem-solving	
12	1	5.0: Data Handling and Probability	5.2: Probability - Expressing probability outcomes as fractions	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for probability as a fraction **Skills:** Express probability outcomes as fractions accurately **Attitudes:** Show understanding of favorable outcomes	**In groups, learners are guided to:** • Identify total possible outcomes • Identify favorable outcomes • Express probability as fraction of favorable to total outcomes • Simplify probability fractions • Calculate probabilities from various scenarios • Solve word problems involving probability • Verify answers	How do we express the chance of an event happening as a fraction?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Colored balls/beads - Bags - Calculator - Pencil - Exercise books	- Observation - Written assignments - Problem-solving tasks	
12	2	5.0: Data Handling and Probability	5.2: Probability - Expressing probability outcomes as fractions	**At the end of the lesson, the learner should be able to:** **Knowledge:** State the formula for probability as a fraction **Skills:** Express probability outcomes as fractions accurately **Attitudes:** Show understanding of favorable outcomes	**In groups, learners are guided to:** • Identify total possible outcomes • Identify favorable outcomes • Express probability as fraction of favorable to total outcomes • Simplify probability fractions • Calculate probabilities from various scenarios • Solve word problems involving probability • Verify answers	How do we express the chance of an event happening as a fraction?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Colored balls/beads - Bags - Calculator - Pencil - Exercise books	- Observation - Written assignments - Problem-solving tasks	
12	3	5.0: Data Handling and Probability	5.2: Probability - Expressing probability as decimals and percentages	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the relationship between probability in fractions, decimals and percentages **Skills:** Convert probability from fractions to decimals and percentages **Attitudes:** Demonstrate proficiency in probability conversions	**In groups, learners are guided to:** • Convert probability fractions to decimals • Convert probability fractions to percentages • Understand that probability in decimals cannot exceed 1 • Understand that probability in percentages cannot exceed 100% • Calculate complementary probabilities • Solve problems in different forms • Apply probability in real contexts	Why is probability sometimes expressed as decimals or percentages?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Calculator - Pencil - Exercise books - Conversion charts	- Observation - Written tests - Problem-solving	
12	4	5.0: Data Handling and Probability	5.2: Probability - Using IT to play probability games	**At the end of the lesson, the learner should be able to:** **Knowledge:** Identify digital tools for probability activities **Skills:** Use technology to play games involving probability and simulate experiments **Attitudes:** Appreciate technology in learning probability	**In groups, learners are guided to:** • Access online probability games • Use software to simulate coin flips • Use apps to simulate dice rolls • Play digital probability games • Record results from digital experiments • Compare manual and digital experiments • Discuss advantages of using technology	How does technology help us understand probability better?	- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Computers/tablets - Internet access - Probability apps/software - Projector - Recording sheets	- Observation - Digital portfolio - Practical demonstration - Oral presentation	

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12	5	5.0: Data Handling and Probability	5.2: Probability - Using IT to play probability games	**At the end of the lesson, the learner should be able to:** **Knowledge:** Explain the key concept in the sub-strand. **Skills:** Apply the concept in mathematical tasks and real-life situations. **Attitudes:** Appreciate the usefulness of mathematics in everyday life.	**In groups, learners are guided to:** • Discuss the key ideas and relate them to familiar situations. • Demonstrate the relevant mathematical procedure using suitable learning resources. • Practise the skills and present findings to peers.				